

cv_all_voting

scalars_cv_allmodels

key	value
n_samples	10481
main_kwargs_applied.substitute	False
main_kwargs_applied.normalize_final_letters	True
timings_seconds.cross_wer	11.5
timings_seconds.ranking	375.8
timings_seconds.model_filtration	8736.2
timings_seconds.rank_distribution	10021.8
ranking_stats.n_models	9
ranking_stats.pool	seamless_m4t+whisper_v3+whisper_v2+fastconformer+xlsr+w2vbert+hubert+wav2vec_sb+cohere
rank_distribution_summary.n_subpools	382
rank_distribution_summary.n_boot_inner	1000
rank_distribution_summary.n_fusions	466
rank_distribution_summary.n_scorings	1971
rank_distribution_summary.seconds_fuse_and_score	1220.6
rank_distribution_summary.seconds_bootstrap	64.6
rank_distribution_summary.weighting	filtration
rank_distribution_summary.exponent	1
rank_distribution_summary.mean_entropy	0.2346
rank_distribution_summary.max_entropy	0.4998
rank_distribution_summary.n_models_modal_differs_from_full	0

ranking_cv_allmodels

model	role	n_samples	wer_pseudo_loo	cer_pseudo_loo	rank_pseudo
cohere	voter	10481	5.813	1.7304	1
fastconformer	voter	10481	7.2542	2.1648	2
hubert	voter	10481	9.4225	2.5478	3
seamless_m4t	voter	10481	9.736	3.3991	4
whisper_v3	voter	10481	12.4314	4.0537	5
whisper_v2	voter	10481	16.277	5.4653	6
xlsr	voter	10481	21.2741	5.9742	7
w2vbert	voter	10481	22.3989	6.1796	8
wav2vec_sb	voter	10481	27.7003	9.1769	9

model_confidence_cv_allmodels

model	rank_pseudo_full	rank_modal	p_modal	p_at_full_rank	entropy_norm	rank_mean	rank_sd	rank_ci_lo	rank_ci_hi	agrees_with_full
cohere	1	1	0.7755	0.7755	0.3225	1.3153	0.7044	1	4	True
fastconformer	2	2	0.6522	0.6522	0.4562	1.9923	0.7712	1	4	True
hubert	3	3	0.5854	0.5854	0.4998	3.1835	0.7391	2	5	True
seamless_m4t	4	4	0.5755	0.5755	0.4811	3.6979	0.7093	2	5	True
whisper_v3	5	5	0.8294	0.8294	0.2626	4.8279	0.4741	3	5	True
whisper_v2	6	6	0.9801	0.9801	0.0525	5.9881	0.1624	6	6	True
xlsr	7	7	0.9913	0.9913	0.0254	6.9985	0.0933	7	7	True
w2vbert	8	8	0.9964	0.9964	0.0108	7.9964	0.0598	8	8	True
wav2vec_sb	9	9	1	1	-0	9	0	9	9	True

rank_distribution_cv_allmodels

model	rank_1	rank_2	rank_3	rank_4	rank_5	rank_6	rank_7	rank_8	rank_9
seamless_m4t	0.0003	0.0541	0.2822	0.5755	0.0863	0.0016	0	0	0
whisper_v3	0	4.0738e-06	0.0279	0.1295	0.8294	0.0132	0	0	0
whisper_v2	0	0	0	0.0022	0.0126	0.9801	0.0051	0	0
fastconformer	0.2132	0.6522	0.0807	0.0368	0.0171	0	0	0	0
xlsr	0	0	0	0	0	0.0051	0.9913	0.0036	0
w2vbert	0	0	0	0	0	0	0.0036	0.9964	0
hubert	0.0109	0.1213	0.5854	0.2383	0.0442	0	0	0	0
wav2vec_sb	0	0	0	0	0	0	0	0	1
cohere	0.7755	0.1724	0.0238	0.0178	0.0105	0	0	0	0

pair_probability_cv_allmodels

pair	pred_winner_full	gap_wer_pseudo	p_first_beats_second	p_supports_full
seamless_m4t hubert	hubert	0.3135	0.2866	0.7134
fastconformer cohere	cohere	1.4412	0.2132	0.7868
fastconformer hubert	fastconformer	2.1684	0.8845	0.8845
seamless_m4t whisper_v3	seamless_m4t	2.6954	0.9073	0.9073
seamless_m4t fastconformer	fastconformer	2.4819	0.0688	0.9312
whisper_v3 hubert	hubert	3.0089	0.0588	0.9412
hubert cohere	cohere	3.6096	0.0467	0.9533
seamless_m4t cohere	cohere	3.9231	0.0429	0.9571
whisper_v3 fastconformer	fastconformer	5.1772	0.0212	0.9788
whisper_v3 whisper_v2	whisper_v3	3.8456	0.9868	0.9868
whisper_v3 cohere	cohere	6.6184	0.0126	0.9874
whisper_v2 xlsr	whisper_v2	4.9971	0.9949	0.9949

pair	pred_winner_full	gap_wer_pseudo	p_first_beats_second	p_supports_full
seamless_m4t whisper_v2	seamless_m4t	6.5409	0.9964	0.9964
xlsr w2vbert	xlsr	1.1248	0.9964	0.9964
whisper_v2 hubert	hubert	6.8544	0.0002	0.9998
w2vbert cohere	cohere	16.5859	0	1
xlsr wav2vec_sb	xlsr	6.4262	1	1
hubert wav2vec_sb	hubert	18.2777	1	1
fastconformer wav2vec_sb	fastconformer	20.4461	1	1
w2vbert hubert	hubert	12.9763	0	1
w2vbert wav2vec_sb	w2vbert	5.3014	1	1
xlsr hubert	hubert	11.8515	0	1
xlsr cohere	cohere	15.4611	0	1
whisper_v2 w2vbert	whisper_v2	6.1219	1	1
fastconformer xlsr	fastconformer	14.0199	1	1
whisper_v2 cohere	cohere	10.464	0	1
whisper_v2 wav2vec_sb	whisper_v2	11.4233	1	1
whisper_v2 fastconformer	fastconformer	9.0228	0	1
whisper_v3 wav2vec_sb	whisper_v3	15.2689	1	1
whisper_v3 w2vbert	whisper_v3	9.9675	1	1
whisper_v3 xlsr	whisper_v3	8.8427	1	1
seamless_m4t wav2vec_sb	seamless_m4t	17.9642	1	1
seamless_m4t w2vbert	seamless_m4t	12.6628	1	1
seamless_m4t xlsr	seamless_m4t	11.538	1	1
fastconformer w2vbert	fastconformer	15.1447	1	1
wav2vec_sb cohere	cohere	21.8873	0	1

model_filtration_cv_allmodels

model	n_subpools	f_bar	f_sd	f_min	f_max	survival
cohere	219	0.0443	0.0122	0.0261	0.099	0.9557
fastconformer	219	0.0679	0.0165	0.0427	0.1342	0.9321
hubert	219	0.0957	0.0226	0.0653	0.1762	0.9043
seamless_m4t	219	0.1213	0.0196	0.0892	0.2082	0.8787
whisper_v3	219	0.1522	0.0399	0.0688	0.2938	0.8478
whisper_v2	219	0.2337	0.0422	0.1355	0.3835	0.7663
xlsr	219	0.3292	0.0587	0.1897	0.4751	0.6708
w2vbert	219	0.387	0.047	0.2874	0.5229	0.613
wav2vec_sb	219	0.4865	0.0426	0.3578	0.5971	0.5135

by_pool_size_cv_allmodels

k	pools	total_weight	mean_entropy	max_entropy
4	126	0.2557	0.0277	0.1301
5	126	0.3196	0.0292	0.1326
6	84	0.2557	0.0232	0.1099
7	36	0.1279	0.0269	0.1327
8	9	0.0365	0.026	0.071
9	1	0.0046	0.0207	0.0207

cross_wer_cv_allmodels

	seamless_m4t	whisper_v3	whisper_v2	fastconformer	xlsr	w2vbert	hubert	wav2vec_sb	cohere
seamless_m4t	0	15.72	19.32	11.73	24.8	25.68	13.77	30.64	10.12
whisper_v3	15.72	0	14.74	14.93	25.47	26.19	17.02	31.68	13.81
whisper_v2	19.32	14.74	0	18.56	28	28.78	20.46	33.97	17.5
fastconformer	11.73	14.93	18.56	0	22.75	24.05	10.29	28.79	7.03
xlsr	24.8	25.47	28	22.75	0	31.64	23.47	31.08	22.24
w2vbert	25.68	26.19	28.78	24.05	31.64	0	25.3	37.17	23.29
hubert	13.77	17.02	20.46	10.29	23.47	25.3	0	29.57	9.2
wav2vec_sb	30.64	31.68	33.97	28.79	31.08	37.17	29.57	0	28.78
cohere	10.12	13.81	17.5	7.03	22.24	23.29	9.2	28.78	0

subpool_filtration_cv_allmodels (first 60 of 1971 rows)

pool	k	model	n_samples	filtered_out	filtered_out_rate
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	seamless_m4t	10481	1626	0.1551
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	whisper_v3	10481	1377	0.1314
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	whisper_v2	10481	2556	0.2439
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	fastconformer	10481	1407	0.1342
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	seamless_m4t	10481	1554	0.1483
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	whisper_v3	10481	979	0.0934
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	whisper_v2	10481	1863	0.1778
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	xlsr	10481	4106	0.3918
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	seamless_m4t	10481	1600	0.1527
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	whisper_v3	10481	961	0.0917
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	whisper_v2	10481	1854	0.1769
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	w2vbert	10481	4508	0.4301
seamless_m4t+whisper_v3+whisper_v2+hubert	4	seamless_m4t	10481	1615	0.1541
seamless_m4t+whisper_v3+whisper_v2+hubert	4	whisper_v3	10481	1360	0.1298

pool	k	model	n_samples	filtered_out	filtered_out_rate
seamless_m4t+whisper_v3+whisper_v2+hubert	4	whisper_v2	10481	2495	0.238
seamless_m4t+whisper_v3+whisper_v2+hubert	4	hubert	10481	1847	0.1762
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	seamless_m4t	10481	1295	0.1236
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	whisper_v3	10481	823	0.0785
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	whisper_v2	10481	1677	0.16
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	wav2vec_sb	10481	5417	0.5168
seamless_m4t+whisper_v3+whisper_v2+cohere	4	seamless_m4t	10481	1611	0.1537
seamless_m4t+whisper_v3+whisper_v2+cohere	4	whisper_v3	10481	1471	0.1403
seamless_m4t+whisper_v3+whisper_v2+cohere	4	whisper_v2	10481	2631	0.251
seamless_m4t+whisper_v3+whisper_v2+cohere	4	cohere	10481	994	0.0948
seamless_m4t+whisper_v3+fastconformer+xlsr	4	seamless_m4t	10481	1272	0.1214
seamless_m4t+whisper_v3+fastconformer+xlsr	4	whisper_v3	10481	1847	0.1762
seamless_m4t+whisper_v3+fastconformer+xlsr	4	fastconformer	10481	824	0.0786
seamless_m4t+whisper_v3+fastconformer+xlsr	4	xlsr	10481	4216	0.4023
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	seamless_m4t	10481	1304	0.1244
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	whisper_v3	10481	1859	0.1774
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	fastconformer	10481	877	0.0837
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	w2vbert	10481	4670	0.4456
seamless_m4t+whisper_v3+fastconformer+hubert	4	seamless_m4t	10481	1633	0.1558
seamless_m4t+whisper_v3+fastconformer+hubert	4	whisper_v3	10481	2622	0.2502
seamless_m4t+whisper_v3+fastconformer+hubert	4	fastconformer	10481	945	0.0902
seamless_m4t+whisper_v3+fastconformer+hubert	4	hubert	10481	1461	0.1394
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	seamless_m4t	10481	1050	0.1002
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	whisper_v3	10481	1726	0.1647
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	fastconformer	10481	654	0.0624
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	wav2vec_sb	10481	5522	0.5269
seamless_m4t+whisper_v3+fastconformer+cohere	4	seamless_m4t	10481	1692	0.1614
seamless_m4t+whisper_v3+fastconformer+cohere	4	whisper_v3	10481	2786	0.2658
seamless_m4t+whisper_v3+fastconformer+cohere	4	fastconformer	10481	1045	0.0997
seamless_m4t+whisper_v3+fastconformer+cohere	4	cohere	10481	567	0.0541
seamless_m4t+whisper_v3+xlsr+w2vbert	4	seamless_m4t	10481	1206	0.1151
seamless_m4t+whisper_v3+xlsr+w2vbert	4	whisper_v3	10481	1414	0.1349
seamless_m4t+whisper_v3+xlsr+w2vbert	4	xlsr	10481	3361	0.3207
seamless_m4t+whisper_v3+xlsr+w2vbert	4	w2vbert	10481	3711	0.3541
seamless_m4t+whisper_v3+xlsr+hubert	4	seamless_m4t	10481	1244	0.1187
seamless_m4t+whisper_v3+xlsr+hubert	4	whisper_v3	10481	1861	0.1776
seamless_m4t+whisper_v3+xlsr+hubert	4	xlsr	10481	4058	0.3872
seamless_m4t+whisper_v3+xlsr+hubert	4	hubert	10481	1092	0.1042
seamless_m4t+whisper_v3+xlsr+wav2vec_sb	4	seamless_m4t	10481	1192	0.1137
seamless_m4t+whisper_v3+xlsr+wav2vec_sb	4	whisper_v3	10481	1498	0.1429

pool	k	model	n_samples	filtered_out	filtered_out_rate
seamless_m4t+whisper_v3+xlsr+wav2vec_sb	4	xlsr	10481	2643	0.2522
seamless_m4t+whisper_v3+xlsr+wav2vec_sb	4	wav2vec_sb	10481	4495	0.4289
seamless_m4t+whisper_v3+xlsr+cohere	4	seamless_m4t	10481	1251	0.1194
seamless_m4t+whisper_v3+xlsr+cohere	4	whisper_v3	10481	1952	0.1862
seamless_m4t+whisper_v3+xlsr+cohere	4	xlsr	10481	4316	0.4118
seamless_m4t+whisper_v3+xlsr+cohere	4	cohere	10481	558	0.0532

subpool_ranks_cv_allmodels (first 60 of 4191 rows)

pool	k	model	rank	count
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	seamless_m4t	5	1000
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	whisper_v3	3	898
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	whisper_v3	4	102
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	whisper_v2	6	1000
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	fastconformer	3	102
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	fastconformer	4	898
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	xlsr	7	998
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	xlsr	8	2
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	w2vbert	7	2
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	w2vbert	8	998
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	hubert	2	1000
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	wav2vec_sb	9	1000
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	cohere	1	1000
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	seamless_m4t	5	1000
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	whisper_v3	3	963
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	whisper_v3	4	37
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	whisper_v2	6	1000
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	fastconformer	2	1000
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	xlsr	7	462
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	xlsr	8	538
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	w2vbert	7	538
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	w2vbert	8	462
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	hubert	3	37
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	hubert	4	963
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	wav2vec_sb	9	1000
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	cohere	1	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	seamless_m4t	5	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	whisper_v3	3	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	whisper_v2	6	1000

pool	k	model	rank	count
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	fastconformer	2	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	xlslr	7	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	w2vbert	8	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	hubert	4	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	wav2vec_sb	9	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	cohere	1	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	seamless_m4t	4	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	whisper_v3	3	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	whisper_v2	6	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	fastconformer	2	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	xlslr	7	999
seamless_m4t+whisper_v3+whisper_v2+hubert	4	xlslr	8	1
seamless_m4t+whisper_v3+whisper_v2+hubert	4	w2vbert	7	1
seamless_m4t+whisper_v3+whisper_v2+hubert	4	w2vbert	8	999
seamless_m4t+whisper_v3+whisper_v2+hubert	4	hubert	5	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	wav2vec_sb	9	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	cohere	1	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	seamless_m4t	5	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	whisper_v3	3	854
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	whisper_v3	4	146
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	whisper_v2	6	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	fastconformer	2	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	xlslr	7	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	w2vbert	8	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	hubert	3	146
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	hubert	4	854
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	wav2vec_sb	9	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	cohere	1	1000
seamless_m4t+whisper_v3+whisper_v2+cohere	4	seamless_m4t	5	1000
seamless_m4t+whisper_v3+whisper_v2+cohere	4	whisper_v3	4	1000
seamless_m4t+whisper_v3+whisper_v2+cohere	4	whisper_v2	6	1000